

Red Zone Risk: Forecasting NFL Player Injury Likelihood

(Proposal)

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ABSTRACT

In this project, I will explore and try to expand on existing technology used to predict and prevent injuries occurring in the National Football League. Over the course of the next few weeks, I will utilize various technologies under the python data science umbrella to accomplish this. By the end, I will have trained a model that can predict what variables and data points most commonly cause injuries, which then can be used for many other kinds of analysis.

1 Introduction

In the NFL, injuries are a major nuisance to teams and fans alike. In the 2024-25 NFL season, 2581 athlete games were missed due to injury- which represented a \$213 million loss for teams (VueMotion [1]). Many strides have been made toward managing and preventing these in the past few decades, however, much analysis is still conducted reactively, relying on hindsight. The goal of this research is to generate proactive analysis to show when players are at risk of injury and begin to explore how it might be mitigated

2 Literature Review

In my initial investigation, I could see that much research has been done in this field. The most relevant I found was a GitHub repository using technologies similar to those I plan to use, although their data source has since been deprecated. Their findings include that certain play types, position players, and temperatures had massive impacts on injury likelihood (GitHub. *FreshOats* [2]). I plan to expand upon their research, using newer technology and data points, and will see if I come to similar conclusions or if I'm able to generate new findings and predict injuries in different ways.

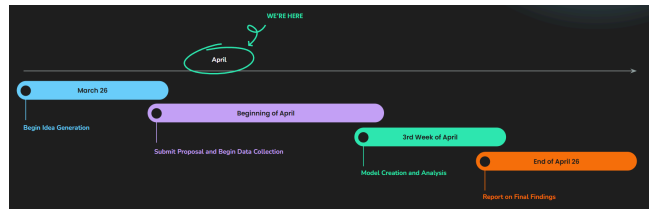
3 Technical Description

In this project, I plan to leverage existing python infrastructure to accomplish my end goals. There is a robust python library called *nlreadpy* that I plan to source most of my data from. It has built in interfacing with *pandas*, which I intend to utilize for many different analytical purposes- including but not limited to data preparation, correlation analysis, and ultimately injury prediction. I also plan to explore using other technologies such as *PyTorch* and *PySpark*, for ML purposes as well as further big data analysis.

4 Evaluation Plan

For this project, I expect the evaluation process to be a little fluid. The datasets I'm looking at span back quite some time, so I think a good way to break down the process would be to train on data from roughly 2017 through 2023, then test against the most recent seasons. In testing, I would hope to be upwards of 80% accurate in predicting injuries, however, I'm not sure how plausible that would be. I also plan to explore other mechanisms for evaluation, such as recall here.

5 Timeline



REFERENCES

- [1] *NFL injuries are a \$213m pain, but they don't have to be: Resource: Vuemotion.* RSS. (n.d.-b). <https://www.vuemotion.com/blog/nfl-injuries-are-a-213m-pain-but-they-dont-have-to-be>
- [2] FreshOats. (n.d.). *FreshOats/NFL_Injury_Analytics: Feature analysis of NFL injuries of the lower body and concussions.* GitHub. https://github.com/FreshOats/NFL_Injury_Analytics